

Your Thesis Title

Your Name

September 30, 2025

Contents

1	Introduction	3
2	Literature Review	4
2.1	Background	4
2.2	Defining Key Concepts	4
2.2.1	DevSecOps	4
2.2.2	ETL	4
2.2.3	Auditability	4
2.2.4	Governance	4
2.2.5	SDLC	4
2.2.6	Data Life Cycle	4
2.3	Research Questions	4
2.3.1	Main Research Question	4
2.3.2	Sub-questions	4
2.4	RQ1: ETL Pipelines in Regulated Industries	4
2.5	RQ2: DevSecOps and Governance in the SDLC	4
2.6	RQ3: Comparing the Data Lifecycle and the SDLC	4
2.7	RQ4: Tooling for Auditability and Governance	4
2.8	Synthesis	4
3	Methodology	5
4	Baseline Pipeline	6
5	Proposed Architecture	7
6	Prototype Implmentation	8
7	Evaluation	9

Chapter 1

Introduction

This is the introduction chapter. Write your introduction here.

Chapter 2

Literature Review

This chapter reviews the state of knowledge relevant to the design of ETL pipelines in regulated industries and the application of DevSecOps principles. It begins by introducing the context of ETL and governance requirements, then defines the key concepts used in this thesis. From this foundation, the research questions are posed explicitly. Each subsequent subsection addresses one sub-question through a focused review of the literature.

2.1 Background

2.2 Defining Key Concepts

2.2.1 DevSecOps

2.2.2 ETL

2.2.3 Auditability

2.2.4 Governance

2.2.5 SDLC

2.2.6 Data Life Cycle

2.3 Research Questions

Drawing from the gaps identified in prior work, this paper will address the following:

2.3.1 Main Research Question

How can a DevSecOps toolchain be designed to support ETL workflows in regulated industries, enhancing auditability and governance?

2.3.2 Sub-questions

1. How does ETL manifest in regulated industries, and what auditability and governance challenges are faced?

2. How do DevSecOps practices enhance auditability and governance within the software development lifecycle?

3. In what ways do the stages of the data lifecycle align with or diverge from the stages of the software development lifecycle, and how does this affect the integration of DevSecOps practices?

4. What are the key categories of tooling in DevSecOps and in ETL pipelines, and how do their features compare in terms of supporting auditability and governance?

2.4 RQ1: ETL Pipelines in Regulated Industries

2.5 RQ2: DevSecOps and Governance in the SDLC

2.6 RQ3: Comparing the Data Lifecycle and the SDLC

2.7 RQ4: Tooling for Auditability and Governance

2.8 Synthesis

Chapter 3

Methodology

Chapter 4

Baseline Pipeline

Chapter 5

Proposed Architecture

Chapter 6

Prototype Implmentation

Chapter 7

Evaluation

Bibliography